

PyTV Module Design Document

Module:	MuL
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Revision History:

Version	Author	Date	Notes
0.1.0	Bolin Li	2024.11.25	Delete all redundant contents and descriptions in MuL Module
0.1.1	Bolin Li	2024.11.29	Completed the doc, major revise

Release Notes:

There are no release notes for any previous versions.

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1 Preliminaries

1.1 Purpose and scope

This module will realize a general purpose **fixed-point Multiplication** operation. It will take two input specified fixed-point value and calculate their Multiplication value in specified fixed-point format.

1.2 Notations

- LSB: Least significant bit.
- MSB: Most significant bit.
- TCPL: Two's complement positive-only format.
- SMGN: Sign-Magnitude format.
- Functional description: A description of the module's combinational behavior.
- Functional model: A (usually QuBLAS) model of the module's behavior.

1.3 Typographic conventions

Visual Cue	Meaning
(Bullet) •	List of important items.
monospace	Codes or commands typed on the keyboard.
MONOSPACE	Constants, parameters.
<i>italic</i>	Variable values, file and directory names.
<italic>	Placeholder text for users to insert values with.
[item]	Optional items.
bold red text	Important notes, warnings, or errors.
bold green text	Indicating success.
green text	Comments or remarks.
blue text	Revised text or keywords.
monospace/blue	URLs, links.
SMALL CAPS	Hardware submodule names.

1.4 Dependencies

1.4.1 Software dependencies

Software	Version
PyTV (verithon)	0.3
PyTU	0.1.0

1.4.2 Submodule dependencies

This part lists all the PyTV submodule dependencies of the module, including the submodule name and version.

Submodule	Version
DELAY	0.1.0
FxMATCH	0.1.0

2 Functionality

2.1 Overview

The MUL Module receives two input values with configurable precisions, then output their multiplication in a configurable precision fixed-point format. Now the module supports quantization modes and overflow modes listed in section 2.2.2 and 2.2.3, respectively. The module also supports optional **negedge reset** and **enable signal**, which can be configured by IF_RST_N and IF_ENABLE parameters, respectively.

2.2 Configurable parameters

Parameter	Range	Default	Description
QU_IN_1	$[\mathbb{N}^+, \mathbb{Z}, \mathbb{B}]$	[9, 3, True]	Data 1 quantization set
QU_IN_2	$[\mathbb{N}^+, \mathbb{Z}, \mathbb{B}]$	[9, 3, True]	Data 2 quantization set
QU_OUT	$[\mathbb{N}^+, \mathbb{Z}, \mathbb{B}]$	[9, 3, 1]	Output quantization set
QU_MODE	All available modes in 2.2.2	QuMode.TRN.TCPL	Output FxMatch Quantize Mode
OF_MODE	All available modes in 2.2.3	OfMode.WRP.TCPL	Output FxMatch Overflow Mode
N_PIPELINES	$\mathbb{N}$	0	Number of pipeline stages
IF_RST_N	$\mathbb{B}$	False	Sign for including negedge reset
	$[\mathbb{B}, \dots]$	[False, ...]	Requires N_PIPELINES > 0
IF_ENABLE	$\mathbb{B}$	False	Sign for including enable signal

2.2.1 Quantization sets

Quantization set describes the quantization format of value, which is originated from a PyTU class QuType. Here is a list of QuType's members:

Member	Range	Description
DWT	$\mathbb{N}^+$	Data Width
FRAC	$\mathbb{Z}$	Decimal Point Offset, in bits
IF_SIGNED	$\mathbb{B}$	Signed or unsigned input format

2.2.2 Quantization modes

Quantization mode refers to the method applied when dealing with precision loss (at LSB). This module supported the following modes:

Mode	Description
QuMode.TRN.TCPL	Truncate all extra LSB in TCPL format
QuMode.TRN.SMGN	Truncate all extra LSB in Sign-Magnitude (SMGN) format, then convert back to TCPL format
QuMode.RND.POS_INF	Round to nearest quantization level, when the value is equidistant between two levels, round to the larger value
QuMode.RND.NEG_INF	Round to nearest quantization level, when the value is equidistant between two levels, round to the smaller value
QuMode.RND.ZERO	Round to nearest quantization level, when the value is equidistant between two levels, round to the value closer to zero

<code>QuMode.RND.INF</code>	Round to nearest quantization level, when the value is equidistant between two levels, round to the value further from zero
<code>QuMode.RND.CONV</code>	Round to nearest quantization level, when the value is equidistant between two levels, round to the even value (LSB = 0)

2.2.3 Overflow modes

Overflow mode refers to the method applied when dealing with overflow (at MSB). This module supported the following modes:

Mode	Description
<code>OfMode.WRP.TCPL</code>	Wrap around in TCPL format
<code>OfMode.SAT.TCPL</code>	Saturate to the maximum value in TCPL format
<code>OfMode.SAT.SMGN</code>	Saturate to the maximum value in SMGN format, then convert back to TCPL
<code>OfMode.SAT.ZERO</code>	Saturate to zero

2.3 External interface

2.3.1 Clock and reset

Synchronous reset and asynchronous release is the default reset behavior. If the module has a different reset behavior, it should be described here.

Signal	Width	Description
<code>i_clk</code>	1	Clock signal (Requires <code>N_PIPELINES > 0</code>)
<code>i_rst_n</code>	1	Reset signal (Requires <code>N_PIPELINES > 0</code>)

2.3.2 Main Module Interface

Signal	Width	Description
<code>i_data_1</code>	<code>QU_IN_1.DWT</code>	Input signal for data 1
<code>i_data_2</code>	<code>QU_IN_2.DWT</code>	Input signal for data 2
<code>o_result</code>	<code>QU_OUT.DWT</code>	Output signal for result
<code>i_ctrl_en</code>	1	Enable signal for the Mul module

2.4 Functional description

Here depicts a functional description of the module.

Algorithm 1: Module function

```
Input: i_data_1
Input: i_data_2
Input: i_ctrl_en
Output: o_result

1 if IF_ENABLE == True then
2   data_1 = i_data_1 && i_ctrl_en;
3   data_2 = i_data_2 && i_ctrl_en;
4 else
5   data_1 = i_data_1;
6   data_2 = i_data_2;

  // Calculate Quantization Set for multiplication from input data
7 DWT_FIX = QU_IN_1.DWT + QU_IN_2.DWT - 1 - QU_IN_1.IF_SIGNED && QU_IN_2.IF_SIGNED;
8 FRAC_FIX = QU_IN_1.FRAC + QU_IN_2.FRAC;
9 SIGN_FIX = QU_IN_1.IF_SIGNED || QU_IN_2.IF_SIGNED;
10 QU_IN_FIX = QuType(DWT_FIX, FRAC_FIX, SIGN_FIX);

  // Multiply
11 if QU_IN_1.IF_SIGNED && QU_IN_2.IF_SIGNED then
12   Sign bit is assigned to data_1's sign bit data_2's sign bit
13   Perform the Multiplication
14 else if QU_IN_1.IF_SIGNED then
15   Sign bit is assigned to data_1's sign bit
16   Perform the Multiplication
17 else if QU_IN_2.IF_SIGNED then
18   Sign bit is assigned to data_2's sign bit
19   Perform the Multiplication
20 else
21   Perform the Multiplication

  // Quantize and handle overflow to get the specified-format final result
22 o_result = result_unfixed.FxMatch(QU_IN = QU_IN_FIX, QU_OUT = QU_OUT, QU_MODE = QU_MODE,
  OF_MODE = OF_MODE);
```

3 Implementation

3.1 Overall architecture

The MUL Module relies on the FxMATCH module to perform the fixed-point multiplication operation. The FxMATCH module takes two input values with configurable precisions, then output their multiplied value in a configurable precision fixed-point format. The MUL Module then applies the specified quantization and overflow modes to the output value.

When N_PIPELINES is 0, the module is a combinational module and the output value is directly generated in the combinational logic. When N_PIPELINES is greater than 0, the module is a pipelined module with N_PIPELINES pipeline stages, and the output value is generated in a pipeline stage.

3.2 Data and control flow

The module is either a combinational module (when N_PIPELINES is 0) or a fully pipelined module (when N_PIPELINES > 0).

3.3 External sequential behavior

3.3.1 Latency analysis

Latency are N_PIPELINES clock cycles for output the result.

3.3.2 Throughput analysis

This module can handle one frame per clock cycle.